PYSPARK CHALLENGES

Challenge 1: Dropping Nulls 🧹

- ✓ Drop any rows that contain at least one null value.
- ✓ Print the number of rows remaining after dropping the nulls.

> ***Instruction:*** *The provided DataFrame* `df` *contains columns 'name', 'age', and 'city'. Drop rows with nulls from this DataFrame.*

Challenge 2: Removing Duplicates 🚫

- ✓ Remove duplicate rows from the DataFrame.
- ✓ Remove any rows that are duplicates based on the 'email' column.

> ***Instruction:*** *The DataFrame* `df` *has columns 'name', 'email', 'country', and 'subscription_date'. Remove duplicates based on the 'email' column.*

Challenge 3: Filtering by Value 🔍

- ✓ Filter the DataFrame to include only rows where the 'category' column is 'Electronics'.
- ✓ Further filter the result to only include products with a 'price' greater than 1000.

> ***Instruction:*** *The DataFrame* `df` *contains 'product_id', 'category', and 'price'. Apply the filters described above.*

Challenge 4: Renaming Columns ✍️

- ✓ Rename the column 'first_name' to 'firstName'.
- ✓ Rename the column 'last_name' to 'lastName'.
- ✓ Rename 'phone_number' to 'phoneNumber'.

> ***Instruction:*** *The DataFrame* `df` *has columns 'first_name', 'last_name', and 'phone_number'. Rename them as instructed.*

Challenge 5: Handling Datatypes 📊

- ✓ Convert the 'transaction_date' column from a string type to a date type.
- ✓ Convert the 'amount' column from a string type to a double type.

> ***Instruction:*** *The DataFrame* `df` *has 'transaction_id', 'transaction_date', and 'amount'. Perform the necessary type conversions.*

Challenge 6: Conditional Filtering 💡

- ✓ Filter the DataFrame to keep only records where the 'status' is either 'Completed' or 'In Progress'.
- ✓ From the filtered data, select only the 'order_id', 'customer_id', and 'status' columns.

> ***Instruction:*** *The DataFrame* `df` *contains 'order_id', 'customer_id', 'status', and 'ship_date'. Apply the filters and selection as described.*

Challenge 7: Removing Specific Rows 🧹

- ✓ Remove all rows where the 'department' column is 'HR'.
- ✓ From the remaining rows, remove any where the 'salary' is less than 50000.

> ***Instruction:*** *The DataFrame* `df` *has 'employee_id', 'department', and 'salary'. Perform the removals as instructed.*

Challenge 8: Filtering with Multiple Conditions ⚖️

- ✓ Filter the DataFrame to include rows where 'country' is 'USA' and 'age' is greater than or equal to 30.
- ✓ Remove duplicate rows based on the 'user_id' column from the filtered data.

> ***Instruction:*** *The DataFrame* `df` *contains 'user_id', 'country', 'age', and 'join_date'. Apply the filters and duplicate removal.*

Challenge 9: Cleaning Text Data 🧼

- ✓ Remove any leading or trailing whitespace from the 'product_name' column.
- ✓ Convert all values in the 'product_name' column to lowercase.

> ***Instruction:*** *The DataFrame* `df` *has 'product_id', 'product_name', and 'price'. Perform the text cleaning on the 'product_name' column.*

Challenge 10: Combining Tasks 🧩

- ✓ Drop rows where the 'sales' column is null.
- ✓ Filter the DataFrame to only include records where the 'region' is 'West' or 'East'.
- ✓ Remove any duplicate rows based on the 'order_id' column.

> ***Instruction:*** *The DataFrame* `df` *has 'order_id', 'sales', 'region', and 'customer_name'. Apply all three steps in sequence.*

CHALLENGES ANSWERS 💡

Challenge 1: Dropping Nulls

```
df = df.dropna()
print("Number of rows after dropping nulls:", df.count())
```

Challenge 2: Removing Duplicates

```
df = df.dropDuplicates(['email'])
```

Challenge 3: Filtering by Value

```
from pyspark.sql.functions import col
df = df.filter((col('category') == 'Electronics') & (col('price') > 1000))
```

Challenge 4: Renaming Columns

```
df = df.withColumnRenamed('first_name', 'firstName') \
       .withColumnRenamed('last_name', 'lastName') \
       .withColumnRenamed('phone_number', 'phoneNumber')
```

Challenge 5: Handling Datatypes

```
from pyspark.sql.functions import col
from pyspark.sql.types import DateType, DoubleType
df = df.withColumn('transaction_date',
col('transaction_date').cast(DateType())) \
       .withColumn('amount', col('amount').cast(DoubleType()))
```

Challenge 6: Conditional Filtering

```
from pyspark.sql.functions import col
df = df.filter(col('status').isin('Completed', 'In
Progress')).select('order_id', 'customer_id', 'status')
```

Challenge 7: Removing Specific Rows

```
from pyspark.sql.functions import col
df = df.filter(col('department') != 'HR').filter(col('salary') >= 50000)
```

Challenge 8: Filtering with Multiple Conditions

```
from pyspark.sql.functions import col
df = df.filter((col('country') == 'USA') & (col('age') >=
30)).dropDuplicates(['user_id'])
```

Challenge 9: Cleaning Text Data

```
from pyspark.sql.functions import col, trim, lower
df = df.withColumn('product_name', lower(trim(col('product_name'))))
```

Challenge 10: Combining Tasks

```
from pyspark.sql.functions import col
df = df.dropna(subset=['sales'])
df = df.filter(col('region').isin('West', 'East'))
df = df.dropDuplicates(['order_id'])
```